Affordance Hard Negatives for Vision–Language Embeddings

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Abstract

Vision–language models are often evaluated on retrieval, but it remains unclear whether they can distinguish a valid object-part affordance caption from a fluent, purpose-swapped hard negative. A diagnostic set from PACO-LVIS is constructed in which Qwen2.5-VL drafts *[Verb] the [part] to [purpose]* positive captions alongside purpose-swapped negatives; image–caption cosine similarity is then scored using CLIP, SigLIP, and Open-VLJEPA. On a held-out set (N=100, mostly automatic captions), frozen accuracies are 0.63, 0.60, and 0.54, respectively. Ranking fine-tuning improves CLIP substantially (0.87 on 500 pairs, 0.91 on 1000) but raises Open-VLJEPA to only 0.64 at both scales, with most attribute and purpose failures unresolved for the open JEPA model. Code: https://github.com/Ketaki1806/Affordance-Benchmark-VLM.

1 Introduction

Vision–language models achieve strong performance on image–text retrieval, yet controlled probes show they often behave like bags of words, matching objects and attributes without reliably binding who did what to whom (Yuksekgonul et al., 2023; Thrush et al., 2022). Hard negatives were proposed as a fix, but early benchmarks were easy to guess with grammar or common-sense priors alone; SugarCrepe argued that distractors and negatives must stay fluent for failures to be meaningful (Hsieh et al., 2023). Winoground is a partial counterexample: its captions reuse the same words specifically to neutralize language priors, yet follow-up work found its difficulty often traces to confounds outside compositional understanding, such as visual grounding of small objects and referential ambiguity (Thrush et al., 2022; Diwan et al., 2022). Here the evaluation design, not just model capacity, determines whether we measure understanding or shortcuts.

This lesson has mostly targeted subject–object swaps and relational structure. I apply it instead to **affordances**: what an object, or one of its parts, is for. The same door handle can open or close a microwave; the same wall can support a mirror hung at eye level, not at the ceiling. This kind of understanding is central to embodied AI and robot manipulation(Luo et al., 2021, 2022), where an agent must recognize which part of an object affords which action, not just what the object is. Pixel-affordance works such as AGD20K and AffordanceCLIP ask *where* an action lands in an image (Luo et al., 2022; Cuttano et al., 2024). Those datasets are not used. Instead, I build on PACO-LVIS part annotations (Ramanathan et al., 2023), and adapt the evaluation protocol from the compositionality literature: one image, one valid caption, one fluent hard negative, ranked by embedding similarity. The difference is where the contrast of captions sits. Rather than shuffling words around, I keep the wording almost identical and change only the purpose or the spatial goal for the same object.

Research question. Given a PACO image with a visible part, can CLIP-, SigLIP-, and Open-VLJEPA-style embeddings select the valid affordance description over a fluent hard negative that flips purpose or location, and does ranking fine-tuning change which failure types improve?

Frozen dual encoders solve this only moderately well, while a JEPA-style model performs close to chance; ranking fine-tuning closes most of this gap for CLIP but leaves Open-VLJEPA's attribute errors largely unresolved.

Contributions.

- A PACO-LVIS affordance-ranking diagnostic with Qwen2.5-VL positives and purpose/location-swapped hard negatives, plus an explicit stance on when a negative counts as grounded.

- Frozen evaluation of CLIP, SigLIP, and Open-VLJEPA. SigLIP as a dual-encoder control so frozen failures are not CLIP only. Ranking fine-tuning of CLIP and Open-VLJEPA at two data scales.

- A failure-mode analysis of which error types fine-tuning fixes for dual encoders vs. a JEPA-style model.

2 Related Work

Hard-negative VL ranking. SugarCrepe / SugarCrepe++ (Hsieh et al., 2023), ARO (Yuksekgonul et al., 2023), Winoground (Thrush et al., 2022; Diwan et al., 2022), and related suites test fluent or templated caption distractors for compositionality and matching. They share my binary ranking shape but are not about affordance purpose on object parts.

Affordance and parts. AGD20K (Luo et al., 2022), OSAD-Net (Luo et al., 2021), and AffordanceCLIP (Cuttano et al., 2024) study pixel-level affordance grounding. PACO (Ramanathan et al., 2023) provides parts and attributes on LVIS (my image source). Fine-grained open-vocabulary detection-style PACO captions swap appearance attributes, not open/close purpose polarity. Classical affordance mask sets ask where an action applies. Human–object interaction datasets such as HICO (Chao et al., 2015, 2018) and HAKE (Li et al., 2019, 2023) classify person–object interactions. Neither is binary purpose ranking of captions for a PACO part.

Vision Language Joint Embedding Predictive Architecture. VL-JEPA (Chen et al., 2025) predicts in embedding space with a frozen vision encoder and text Y-encoder. **Open-VLJEPA** is evaluated, an open re-implementation that mirrors the stack at much smaller pretraining scale. Hence, these results bound the open checkpoint, not Meta's closed-scale system.

To the best of my knowledge, no existing benchmark combines these three elements, that is, part-level affordance, purpose contrast, and fluent hard negatives, which motivates my interest.

3 Experimental Setup

Data Generation. PACO-LVIS is used because its images typically foreground one object and a visible part, matching the embodied-AI setting where an agent must act on a specific part rather than the whole scene (Figure 1). Captions follow *[Verb] the [part] to [purpose]*; hard negatives preserve fluent syntax but flip purpose (open/close) or spatial goal (wall/ceiling). A pilot of 20 images uses human keep/edit/drop on Qwen's drafts; based on pilot errors, the captioning prompt is revised to reduce part-mismatch and purpose-confusion. The scaled-up N=100 eval set is mostly automatic and unverified, so reported accuracies are an upper-noise bound: the benchmark can inherit captioner mistakes. Fine-tune pools of 500 and 1000 held-out images use the same generation settings; evaluation is always on the same val N=100.

What counts as a grounded negative. A negative that is hard for CLIP is not necessarily well grounded in the image. CLIP-based adversarial filtering to drop easy distractors was considered and rejected, since using CLIP to both select and score negatives is circular. Qwen proposes candidate pairs only; humans filter them at the pilot stage, but not at scale. Affordance is scoped narrowly: physically plausible uses of the named part in its *visible* state, not task- or goal-conditioned uses. PACO supplies object, part, and attributes but not the verb, so Qwen (and pilot annotators) still choose a purpose fitting that part's state. Holding object and part fixed across positive and negative, and varying only purpose or location, *reduces the risk that ranking collapses to object identity*. This does not fully isolate affordance understanding from linguistic similarity. For this reason, text-only closeness and fluency tags are also reported.

Models. **Captioner**: Qwen2.5-VL-7B-Instruct (Bai et al., 2025), chosen for Hugging Face tooling and VRAM fit, not as a size ablation.
Scorers: CLIP ViT-L/14 (Radford et al., 2021), SigLIP SO400M (Zhai et al., 2023) (frozen control in the dual-encoder family), and Open-VLJEPA (dion-jy, 2025), a community reimplementation of the VL-JEPA architecture (Chen et al., 2025) combining a frozen V-JEPA 2 vision encoder (Assran et al., 2025) with Llama-3.2 (Grattafiori et al., 2024) and EmbeddingGemma (Schechter Vera et al., 2025) text stacks.
For CLIP and SigLIP the comparison is standard image–caption cosine. Open-VLJEPA is scored the same way in the shared embedding space, the predictor maps the image and the query to $\hat{y}$, the Y-encoder embeds each caption, and picks the caption with higher cosine to $\hat{y}$.



Figure 1: Project pipeline: PACO-LVIS dataset parts used by Qwen2.5-VL for positive/hard-negative captions. Human pilot check on 20 images. CLIP / SigLIP / Open-VLJEPA scoring. Analysis with error tags and XAI. Val on N=100, ranking FT on 500 and 1000 pairs, occlusion vision share, modality gap, word heatmaps.

Metric: Binary accuracy (correct iff $\cos(I,c_+) > \cos(I,c_-)$) measures how often the model prefers the positive caption. Mean confidence gap $|\cos_+ - \cos_-|$ measures how decisively the model separates the two options, regardless of direction.

Fine-tuning. Adapted CLIP and Open-VLJEPA with a pairwise ranking loss on each training triple (I,c_+,c_-):

$$\mathscr{L} = \mathrm{ReLU}\big(m - \mathrm{sim}_+ + \mathrm{sim}_-\big), \qquad m{=}0.05, \tag{1}$$

where $\mathrm{sim}_\pm$ is the same cosine used at evaluation (Faghri et al., 2018). Training runs for 5 epochs on the held-out 500 and later 1000 image pools. For CLIP, both the vision and text encoders were updated (learning rate 1×10^{-5}). For Open-VLJEPA, the X-encoder remains frozen and the predictor (1×10^{-5}) is trained with a light Y-encoder learning rate (5×10^{-7}). SigLIP is kept frozen and is only used as the dual-encoder control for CLIP.

Analyses. Wrong predictions receive one of three heuristic failure tags: *attribute* (purpose/action confusion), *spatial* (location/part confusion), or *fluency* (caption noise). Text-only Y-space closeness is measured separately, using EmbeddingGemma (Schechter Vera et al., 2025) to compute $\cos(c_+,c_-)$ between positive and negative captions.

Two further metrics are reported, each targeting a different question. The first, occlusion vision share, estimates how much a model's decision relies on the image versus the caption:

$$\text{vision_share} = \frac{\max|\text{grid}|}{\max|\text{grid}| + \max|\text{text}|} \tag{2}$$

It is computed from a leave-one-out text ablation combined with a 3×3 image blackout, applied to the decision margin $\Delta = s_+ - s_-$. The second, embedding modality gap (Liang et al., 2022), measures alignment geometry directly as $\|\bar{z}_{\text{img}} - \bar{z}_{\text{txt}}\|$,

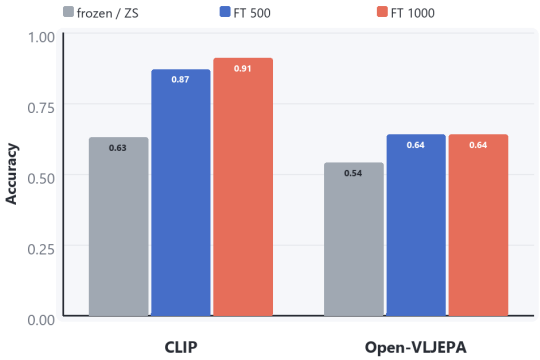


Figure 2: Comparison of frozen / zero-shot accuracy against ranking fine-tuning at two data scales on the same held-out val N=100.

the distance between mean image and text embeddings.

These two metrics answer different questions, decision sensitivity versus representation geometry, and are reported separately rather than merged into a single "grounding" score.

4 Results

4.1 Main accuracies

Table 1 and Figure 2 answer the research question for frozen models: dual encoders sit in the low 60s, while Open-VLJEPA ZS is weaker. SigLIP tracks CLIP closely, so the moderate frozen score is not a CLIP-only quirk. Fine-tuning helps CLIP substantially but raises Open-VLJEPA only to 0.64. Doubling the fine-tuning data (500 to 1000) pushes CLIP further, to 0.91, but leaves Open-VLJEPA flat. Its training loss was already near zero by epoch 5. This pattern points to a capacity or pretraining ceiling under a frozen X-encoder, rather than a simple need for more of the same Qwen-generated pairs.

On the human pilot after SugarCrepe-style edits, CLIP-FT reaches 0.90 (18/20), while VLJEPA-FT falls to 0.45 (9/20). This suggests that cleaner,

Model	Acc.	Notes
CLIP frozen	0.63	gap ≈ 0.021
SigLIP frozen	0.60	dual-encoder control
Open-VLJEPA ZS	0.54	open re-impl
CLIP FT (500)	0.87	ranking FT
Open-VLJEPA FT (500)	0.64	modest lift
CLIP FT (1000)	**0.91**	91/100
Open-VLJEPA FT (1000)	0.64	no further gain

Table 1: Held-out val N=100 with automatic Qwen captions. Fine-tune (FT) pools are the number of train images used for ranking adaptation; evaluation is always on the same val set. SigLIP is frozen only, no FT.

Model	Attribute	Spatial	Fluency
CLIP FT_{500}	21/24 (88%)	7/8 (88%)	4/5 (80%)
CLIP FT_{1000}	21/24 (88%)	7/8 (88%)	4/5 (80%)
Open-VLJEPA FT_{500}	8/38 (21%)	3/5 (60%)	1/3 (33%)
Open-VLJEPA FT_{1000}	9/38 (24%)	4/5 (80%)	1/3 (33%)

Table 2: Failure-tag fix rates among frozen/ZS wrongs after ranking FT (val N=100). Entries are #fixed / #frozen-wrong; see text for why CLIP's rows match despite different overall accuracy.

better-grounded negatives remove leaky cues the open model had come to rely on. *It is exactly the risk automatic hard negatives carry: they can look hard without being well grounded.*

The mean Y-space closeness, $\cos(c_+, c_-)$=0.815 (55% above 0.8), confirms that negatives are close in text space by design: **ranking is not meant to be solvable from language alone**.

4.2 What fine-tuning fixes

Frozen wrongs are attribute-dominated (CLIP 65%, SigLIP 75%, VLJEPA ZS 83% of wrongs), out of 37 total ZS-wrong pairs for CLIP and 46 for Open-VLJEPA. Table 2 shows how many of those wrongs ranking FT later corrects. Figure 3 compares ZS to FT fix rates among those wrongs at both training scales.

CLIP recovers most attribute and spatial wrongs and is broad across all three tags. Note that CLIP's per-tag fix counts are identical at both FT scales (32/37 ZS-wrong examples fixed either way, on different IDs within each tag); the additional 4-point accuracy gain from 500 to 1000 pairs comes from fewer regressions on originally-correct pairs (8 vs. 4), not from correcting more of the tracked ZS failures.

Open-VLJEPA gains little when scaling from 500 to 1000, especially on attribute/purpose failures. Open-VLJEPA's +10 points come from only a thin slice of attribute cases; **29–30/38** attribute wrongs remain uncorrected, which explains the flat result at 1000 pairs.

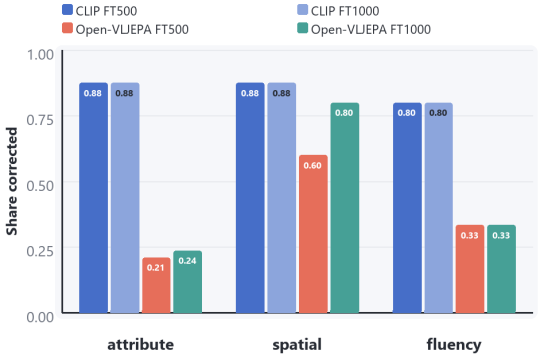


Figure 3: Share of zero-shot wrong predictions corrected by ranking FT, by failure tag (val N=100).

4.3 Qualitative examples

Attribute. Microwave (`lvis_486018`):

> \+ Open the microwave by pulling the door handle.
> − Close the microwave by pulling the door handle.

Frozen CLIP, SigLIP, and Open-VLJEPA all prefer the close caption, the intended purpose-polarity swap. Figure 4 shows why: CLIP's word heatmap for *microwave* lands on the blank wall rather than the appliance, and the CLIP and Open-VLJEPA 3×3 occlusion grids for the same pair are stacked below it.

Spatial. Mirror (`lvis_75183`):

> \+ Hang the mirror on the wall.
> − Hang the mirror on the ceiling.

CLIP and SigLIP fail this ranking, illustrating layout blindness of global embeddings. Figure 5 stacks CLIP heatmaps for *wall* (sensible) and *ceiling* (hot spots on wall/mantel).

Fluency. Pillow image (`lvis_422959`):

> \+ "Press the button to activate the lamp."
> − "Turn the switch to turn off the lamp."

Both captions describe a lamp, not the pillow, signaling an object mismatch. The wrong ranking here is caption noise, not a model failure. Cases like this mean the mostly automatic N=100 set is noisy: reported accuracies should be read as an approximate diagnostic under imperfect captions, not as scores on a fully human-validated set.

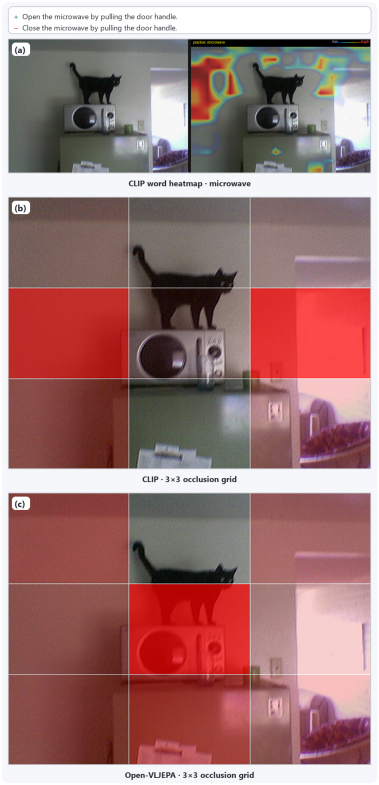


Figure 4: Attribute hard-negative on PACO `lvis_486018`. Caption strip: Qwen positive (open) vs. negative (close); all three frozen scorers prefer close. (a) CLIP word heatmap for *microwave*: heat on blank wall, not the appliance; (b) CLIP 3×3 occlusion grid (Δ-drop); (c) Open-VLJEPA grid: stronger vision response on the object cell, still a ranking failure.



Figure 5: Spatial hard-negative on PACO `lvis_75183`. Caption strip: Qwen positive (wall) vs. negative (ceiling); CLIP and SigLIP fail. (a) CLIP heatmap for *wall* on wall surfaces around the mirror (b) CLIP heatmap for *ceiling* stays on wall/mantel, layout blindness under a fluent spatial distractor.

4.4 Secondary Diagnostics: Vision Reliance and Modality Gap

On N=100, mean occlusion vision share is ≈0.23–0.26: the binary decision is mostly text-explained under leave-one-out / coarse blackout (Figure 6). Embedding modality gap (Liang et al., 2022) is large for CLIP/SigLIP (≈ 0.97 / 1.04) but small for Open-VLJEPA (≈ 0.27) with higher matched cosines, yet *better alignment does not yield better affordance ranking here* (Figure 7). Taken together, these diagnostics show that *low vision reliance and tight embedding alignment are properties of a model's representation space, not guarantees of correct affordance ranking. Neither metric substitutes for the task accuracy* reported in Table 1.



Figure 6: Occlusion vision share on val N=100 decision sensitivity measured by vision share (2), from leave-one-out text ablation and 3×3 image blackout on $\Delta = s_+ - s_-$. Bars are means over all / correct/wrong pairs; *vision is only 0.23–0.26 of peak sensitivity*.

5 Conclusion

This paper asked whether vision–language embeddings can rank PACO part affordances against fluent, purpose-swapped hard negatives, and whether ranking fine-tuning changes which failure types improve. Frozen dual encoders reach only 60–63%

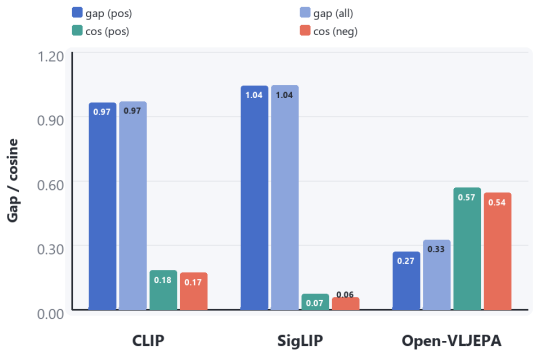


Figure 7: Embedding modality gap on val N=100 (alignment geometry): $\|\bar{z}_{\text{img}} - \bar{z}_{\text{txt}}\|$ and matched cosines. CLIP/SigLIP show a large gap and low matched cosines; Open-VLJEPA is far more aligned, without a matching gain on affordance ranking.

accuracy, and Open-VLJEPA is weaker still in the zero-shot setting. Ranking fine-tuning strongly improves CLIP, reaching 0.91 at 1000 training pairs by correcting most attribute and spatial errors, while Open-VLJEPA plateaus at 0.64: its attribute failures largely persist, and adding more Qwen-generated pairs of the same kind does not close that gap.

The design lesson from the human pilot still holds: a hard negative is not automatically a grounded one. Scaling up captioning automatically can inherit the captioner's own mistakes, and a negative is only meaningful when it stays anchored to the same visible part and state as its positive.

This study has several limits. The N=100 evaluation set is mostly automatic and not fully human-verified. CLIP's fine-tuning gains may be partly inflated by Qwen-specific caption patterns rather than affordance understanding alone. Open-VLJEPA is an open reimplementation that is architecture-close but scale-far from Meta's original VL-JEPA, so these results bound the open checkpoint rather than the closed-scale system. Future work should human-edit the full evaluation set and, for the JEPA-style model, prioritize harder, human-written negatives over additional automatically generated pairs of the same kind.

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A Additional Experimental Details

Caption style. Positives follow the template *[Verb] the [part] to [purpose]*. Hard negatives keep similar wording and length but flip purpose polarity (e.g., open/close) or spatial goal (e.g., wall/ceiling). Generation uses Qwen2.5-VL-7B-Instruct with length bounds of 5–10 words and 25–55 characters, a maximum positive–negative length gap of 10 characters, and JSON schema validation with up to three retries (`configs/config.yaml`). Prompt text lives in `configs/prompt_template.txt` and is assembled in `src/`.

Compute. Caption generation, frozen evaluation, ranking fine-tuning, and attribution jobs run on Saarland University LST HTCondor GPU nodes. CLIP and Open-VLJEPA fine-tuning each use a single GPU. The 1000-image train caption pool is produced with sharded Qwen jobs and merged afterward; evaluation always uses the same held-out val set (N=100).

Fine-tuning and checkpoints. Ranking fine-tuning uses hinge loss with margin m=0.05 for 5 epochs on held-out pools of 500 and 1000 pairs (Section 3). CLIP starts from `openai/clip-vit-large-patch14`; reported train-1000 tables use checkpoint `finetuned_affordance_train1000_ep5.pt`. Open-VLJEPA is initialized from the community open checkpoint `best.pt` and fine-tuned to the matching `finetuned_affordance_train1000_ep5.pt` (bf16; fp16 caused NaNs with EmbeddingGemma). Train-500 runs write analogous `finetuned_affordance_ep5.pt` files. Checkpoint paths and eval overlays are in the repository configs. https://github.com/Ketaki1806/Affordance-Benchmark-VLM.

B Disclaimer: Use of AI

I used AI coding assistants to scaffold Condor wrappers, plotting helpers, and routine refactors. All experimental numbers come from pipeline runs on LST Condor (and local post-processing of those outputs). Tables were checked against the corresponding eval JSON summaries; citations were not invented by the assistant, and are self-verified. I remain responsible for the scientific claims, figure choices, and final wording.